

## Education

**University of Utah**  
*B.S. Computer Science*

**Salt Lake City, UT**  
*August 2020 – May 2025*

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

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## Skills

**Languages:** Python, TypeScript, C++ , SQL, Java

**Software:** Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

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## Experience

**L3Harris Technologies**  
*Associate Software Engineer*

**Dallas, TX**  
*June 2025 – Present*

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

**Doxy.me**  
*Software Engineer*

**Charleston, SC**  
*August 2024 – May 2025*

- Built a real-time data ingestion pipeline streaming live WebRTC session data (audio, video, transcripts) into ML inference infrastructure on SageMaker and Fargate, supporting sub-second predictions at production scale.
- Designed the schema and processing layer for structured clinical signal extraction, applying RL feedback loops over physician-validated outcomes to continuously refine model accuracy.
- Standardized CI/CD and deployment pipelines across the ML stack, enabling reproducible infrastructure provisioning and zero-downtime releases.

**University of Utah**  
*Undergraduate Research Assistant – FuTURES Lab*

**Salt Lake City, UT**  
*May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

**HEXstream**  
*Software Engineering Intern*

**Chicago, IL**  
*May 2022 – Aug 2022*

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
  - Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.
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## Projects

**University Course Planning/Review Platform** (*Class Best*): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

**Telehealth Mental Health Diagnosis Tool (Doxy.me)**: Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

**Configuration Fuzzer**: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.